VISHAAL A K

Cybersecurity Research Intern | B.E. Computer Science and Engineering, Kumaraguru College of Technology
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PROFESSIONAL SUMMARY

Fourth-year Computer Science Engineering student with hands-on experience in machine learning, cybersecurity, and backend development. Built ML models, security-focused applications, and real-time detection systems using Python and modern backend technologies. Currently contributing to cybersecurity research at VIT-AP University's Centre of Excellence in Cyber Security, with a strong interest in developing secure and scalable AI-driven solutions.

EDUCATION

B.E., Computer Science and Engineering — Kumaraguru College of Technology, Coimbatore *2023 – 2027*
Coursework: Ethical Hacking, Network Security, Cryptography, Operating Systems, Computer Networks, ML Fundamentals
CGPA: 8.02 / 10.0

Higher Secondary Education (Class XII) — Adharsh Vidhyalaya Matric Hr Sec School, Tirupur *2021 – 2023*
Score: 92.70%

TECHNICAL SKILLS

- **Languages:** Python, Java, C, JavaScript (ES6+)
- **Core CS:** Data Structures and Algorithms, Object-Oriented Programming (OOP), System Design, Distributed Systems
- **Web and Backend:** MERN Stack, FastAPI, REST APIs, WebSockets
- **Infrastructure and Tools:** Docker, Apache Kafka, Redis, Git, Linux, Elasticsearch, Wireshark, Nmap, Burp Suite
- **Security:** Secure Coding, Input Validation, OWASP Top 10, Secrets Management, File Upload Security, Prompt Injection Mitigation, RSA Cryptography, TCP/IP
- **Machine Learning and Data:** Scikit-learn, NumPy, Pandas

PROJECTS

E-Commerce Price Prediction System *MERN Stack, Python, Machine Learning*
- Architected a **full-stack price prediction application** integrating a Python ML regression model with a Node.js/Express REST API and a React frontend, enabling real-time price forecasting with sub-500ms response latency.
- Trained and deployed a regression model on 10,000+ product records, achieving a mean absolute error (MAE) of Rs. 45 on a price range of Rs. 100 to Rs. 5,000.
- github.com/Vishaal-15/Ecommerce-Price-Prediction

Intelligent Crop Recommendation System *Python, Scikit-learn, Pandas*
- Built a **machine learning classification model** using Python, Scikit-learn, and Pandas to recommend optimal crops based on soil nutrients (N, P, K) and environmental conditions, achieving 88% prediction accuracy on the test set.
- Compared three algorithms (Logistic Regression, Decision Tree, Random Forest), selecting the best-performing model through cross-validation, improving baseline accuracy by 12%.
- github.com/Vishaal-15/Crop-Recommendation-ML

Secure Prompt-Based Scoring Application *Python, Streamlit*
- Developed a **secure Prompt-Based Scoring application** using Python, Streamlit, and the OpenAI API, applying privacy-by-design principles from the ground up. Reduced the application's attack surface through allowlist-based input filtering, blocking approximately 95% of malicious test inputs.
- Enforced **7 distinct security controls** across the stack, including secrets management, input validation, MIME-type and file-size validation, LLM output validation, secure error handling, data minimization, and HTTPS enforcement, to harden the app against common application security risks.
- github.com/Vishaal-15/prompt-accuracy-scorer

RSA Encryption Implementation *Python, PyCryptodome*
- Engineered **RSA cryptography two ways** in Python: built raw RSA from scratch (manual prime generation, modular inverse, and key derivation) to master the underlying number theory, then developed a production-grade **2048-bit RSA-OAEP** scheme using PyCryptodome for real-world semantic security against chosen-plaintext attacks.
- Validated the implementation against standard cryptographic test vectors, confirming correct modular arithmetic and key exchange behaviour across 100+ test cases, completed within a 3-day self-directed sprint.
- github.com/Vishaal-15/RSA-implementation

EXPERIENCE

Networking and Cloud Intern — Juniper Networks (HPE) *Oct 2024 – Dec 2024*

- Analysed enterprise networking architecture and applied routing configurations across 2 core protocols (BGP, OSPF) in a simulated environment, developing hands-on proficiency in Linux-based network administration and cloud networking fundamentals.
- Configured and tested virtual network topologies, gaining practical exposure to switch and router management and cloud-networking integration across hybrid infrastructure.

Web Development Intern — MotionCut *Nov 2024 – Dec 2024*

- Optimised frontend UI components for responsiveness and load performance, reducing page render time by approximately 30% through CSS restructuring and asset optimisation.
- Collaborated with the development team on feature delivery across 3 web modules, following agile sprint cycles and version control best practices using Git.

Security Research Intern — VIT-AP University *Jun 2026 – Present*

- Built a high-performance, production-ready Extended Detection & Response (XDR) platform capable of scale-testing 100,000+ endpoints, leveraging an async FastAPI backend, Redis caching, and Elasticsearch log aggregation for real-time security telemetry.
- Engineered a resilient async backend processing tier utilizing Elasticsearch for high-speed log indexing and paginated search, and integrated an automated response orchestrator featuring MITRE ATT&CK mapping, SLA-enforced incident ticketing (Jira/ServiceNow), and Kasm Remote Browser Isolation.

CERTIFICATIONS AND ACHIEVEMENTS

- **Foundations of Cybersecurity** — Google / Coursera: industry-recognised certification covering threat analysis, security frameworks, and risk management.
- **Cryptography and Network Security** — Kumaraguru Institutions: formal coursework certification in encryption standards, PKI, and secure protocols.
- **AWS Cloud Architect** — Amazon Web Services: certification covering cloud architecture, compute, storage, and networking fundamentals on AWS.
- **Introduction to Information Security** — Great Learning
- **Introduction to Operating Systems: Virtualization** — Codio
- Secured 92.7% in Higher Secondary Education (2023), top academic performance at school level.

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

- Participated in hackathons focused on cybersecurity and AI, representing the college in inter-collegiate events.
- Engaged in peer knowledge-sharing sessions on cryptography and network security concepts, supporting junior students within the CSE department.